Thm. Every non-constant analytic function is open map.

Proof. Suppose that $f: D \to \mathbb{C}$ is analytic on the connected open set D.
Let $\mathcal{U} \subseteq D$ be an open set. Given $z_0 \in \mathcal{U}$, choose $\delta > 0$ such that $\overline{B_\delta(z_0)} \subseteq \mathcal{U}$ and $f(z) - f(z_0) \neq 0$ for all $|z - z_0| = \delta$. (It is possible by the identity theorem.)
Choose $m = \min\{|f(z) - f(z_0)| \mid |z - z_0| = \delta\}$, by the min-max theorem.
Let $w_1 \in \{w \in \mathbb{C} \mid |w - f(z_0)| < m\}$. Then, on $|z - z_0| = \delta$,

$$|w_1 - f(z_0)| < m \leq |f(z) - f(z_0)|.$$

By the Rouche's theorem,

$$[f(z) - f(z_0)] + [f(z_0) - w_1] = f(z) - w_1$$

has same zeros in $|z - z_0| < \delta$ to $f(z) - f(z_0)$. So, $f(z) - w_1 = 0$ has at least one zero,
that is, $f(z) = w_1$ at least once in $|z - z_0| < \delta$.
Since w_1 was arbitrary, $\{w \in \mathbb{C} \mid |w - f(z_0)| < m\} \subseteq f[B_\delta(z_0)]$.